

CSA1402 - COMPILER DESIGN

Title: Compiler vs Interpreter

Presented by

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Introduction

What is a Language Translator?

A language translator converts a high-level programming language into machine language so that the computer can understand and execute it.

There are three types of language translators:

- **Compiler**
- **Interpreter**
- **Assembler**

Example Languages

- **Compiler** → C, C++, Go
- **Interpreter** → Python, JavaScript, Ruby

What is a Compiler?

A compiler translates the entire source program into machine code before execution.

Features

- Converts the whole program at once.
- Generates an executable (.exe) file.
- Detects errors after compilation.
- Execution is very fast.
- Suitable for large applications.

Examples

- GCC (GNU Compiler Collection)
- Clang
- Microsoft Visual C++



What is an Interpreter?

An interpreter translates and executes the program line by line.

Features

- No executable file is created.
- Stops immediately if an error occurs.
- Easier to debug.
- Slower execution.
- Good for scripting and testing.

Examples

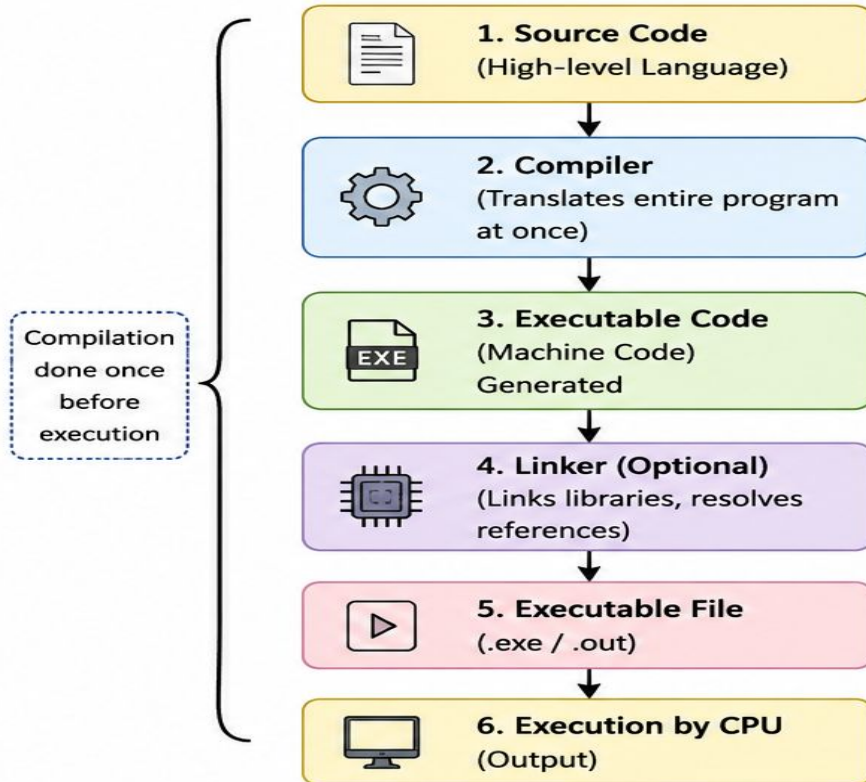
- Python Interpreter
- Node.js (JavaScript)
- Ruby Interpreter



Compiler vs Interpreter

Feature	Compiler	Interpreter
 Translation	Translates entire program	Translates line by line
 Execution Speed	Faster execution	Slower execution
 Executable File	Generates executable file	No executable file
 Error Reporting	Errors shown after compilation	Errors shown immediately
 Used For	Used for C, C++	Used for Python, JavaScript

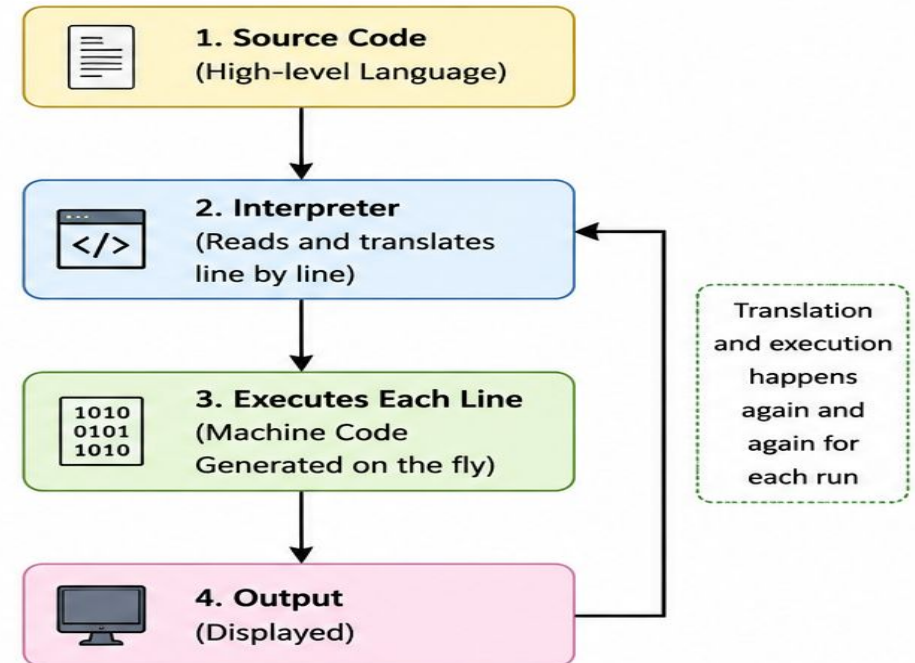
Execution Flow of Compiler



Compiler Flow Summary

- Entire program is translated at once.
- Generates an executable file.
- Execution is fast because translation is done only once.

Execution Flow of Interpreter



Interpreter Flow Summary

- Program is translated and executed line by line.
- No executable file is generated.
- Translation happens every time the program runs.

example code in C:

```
#include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);

    if (n % 2 == 0)
        printf("Even");
    else
        printf("Odd");

    return 0;
}
```

C Source Code (.c)



Compiler (GCC)



Executable File (.exe)



Run Executable



Output (Even/Odd)

Example code in python:

```
# Program to check whether  
a number is odd or even
```

```
number = int(input("Enter an integer: "))
```

```
if number % 2 == 0:
```

```
    print(f" {number} is an Even number.")
```

```
else:
```

```
    print(f" {number} is an Odd number.")
```

Python Source Code (.py)



Python Interpreter



Reads One Line



Executes That Line



Reads Next Line



Output (Even/Odd)



COMPILER

Advantages



1. **Faster Execution** – Compiled programs run faster because the entire program is translated before execution.



2. **Creates Executable File** – Generates a standalone executable (.exe) that can be run without the compiler.



3. **Better Performance** – Optimizes the code for efficient use of CPU and memory.



4. **Suitable for Large Applications** – Best for developing operating systems, games, and desktop software.

Disadvantages



1. **Compilation Takes Time** – The entire program must be compiled before it can run.



2. **Difficult to Debug** – Errors are reported after compilation, making debugging less convenient.



3. **Platform Dependent** – Executable files are usually created for a specific operating system or processor.



4. **Requires More Memory** – Compilation and executable files consume more storage and memory.



INTERPRETER

Advantages



1. **Easy to Debug** – Errors are displayed immediately when encountered.



2. **No Compilation Required** – Executes the program directly without creating an executable file.



3. **Platform Independent** – The same source code can run on different platforms if the appropriate interpreter is installed.



4. **Faster Development** – Suitable for testing, scripting, and rapid application development.

Disadvantages



1. **Slower Execution** – The program is translated and executed line by line.



2. **No Executable File** – An interpreter is required every time the program runs.



3. **Less Efficient** – Translation happens each time the program is executed, reducing performance.



4. **Not Suitable for Large Applications** – Slower execution makes it less ideal for performance-critical software.

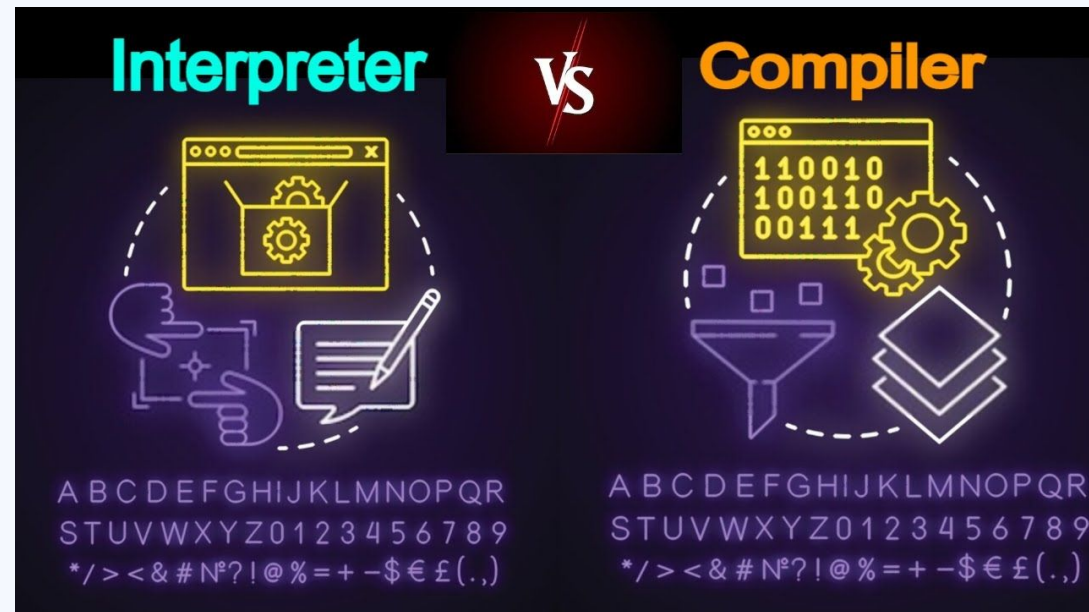
Where Are They Used?

Compiler

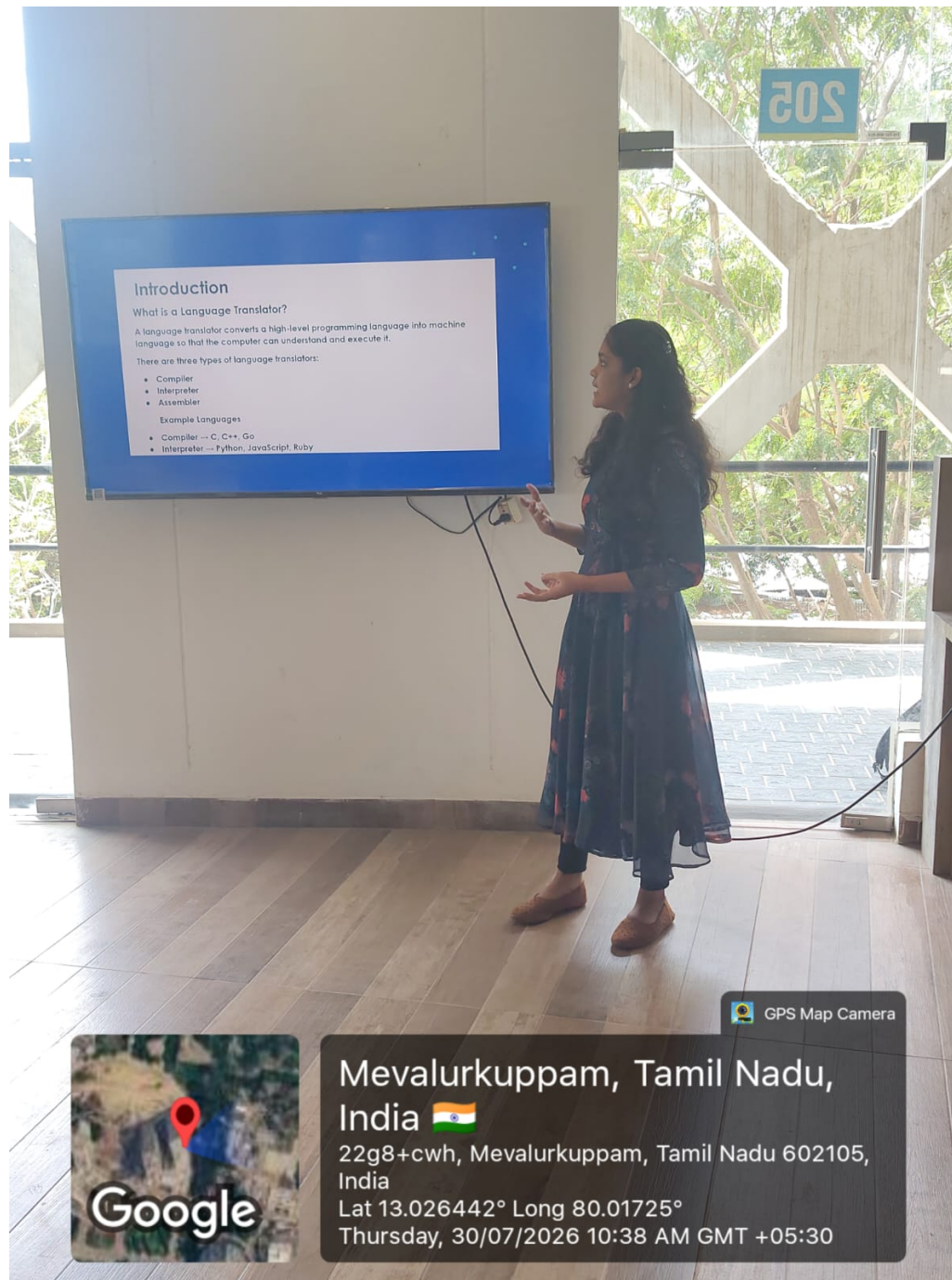
- Operating Systems
- Games
- Desktop Applications
- Embedded Systems

Interpreter

- Web Development
- Data Science
- Artificial Intelligence
- Automation Scripts



THANK YOU!



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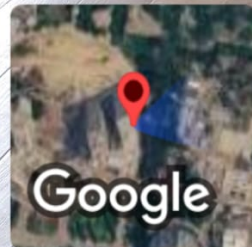
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GPS Map Camera



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